Peter Dietrich
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August 23, 2022
NRC-22-0029

10 CFR 50.73

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, DC 20555-0001

Fermi 2 Power Plant
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2022-003

Pursuant to 10 CFR 50.73(a)(2)(iv)(A), DTE Electric Company (DTE) is submitting LER No. 2022-003, Turbine trip and subsequent reactor trip due to mayflies.

No new commitments are being made in this submittal.

Should you have any questions or require additional information, please contact Mr. Eric Frank, Manager – Nuclear Licensing, at (734) 586-4772.

Sincerely,

Peter Dietrich
Senior Vice President and Chief Nuclear Officer

Enclosure: Licensee Event Report No. 2022-003, Turbine trip and subsequent reactor trip due to mayflies.

cc: NRC Project Manager
NRC Resident Office
Regional Administrator, Region III

Enclosure to
NRC-22-0029

Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43

Licensee Event Report (LER) No. 2022-003
Turbine trip and subsequent reactor trip due to mayflies

NRC FORM 366 (08-2020) U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023

LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
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4. Title
Turbine trip and subsequent reactor trip due to Mayflies

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
06	24	2022	2022	- 003 -	00	08	23	2022	N/A	05000
									N/A	05000

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Eric Frank – Manager, Nuclear Licensing	734-586-4772

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
C	EL	INS	LAPP	Yes					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☐ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

At 2338 EDT on June 24, 2022, while in Mode 1, 100 percent power, Fermi 2 experienced a turbine trip followed by a reactor SCRAM, due to differential relaying sensing a generator output fault. The generator output fault occurred when a swarm of mayflies caused an arc from a 345 kV bus bar to the bottom of the insulator during a hatch. The National Weather Service data indicated the Mayfly hatch that occurred near Fermi 2 site on June 24, 2022 was significant, and compensatory actions were taken, however, additional subsequent actions were taken to minimize the potential for future switchyard operations being impacted from mayflies. Area and building lighting has been reduced further to limit attracting mayflies to sensitive plant equipment and the mayfly infestation buildup and forecast will be closely monitored as part of the mayfly mitigation plan. The seasonal mayfly control procedure has been updated to reflect these changes. A 4-hour event notification (EN 55964) was made to the NRC pursuant to Title 10 Code of Federal Regulations (10 CFR) 50.72(b)(2)(iv)(B) for RPS Actuation (Reactor Scram) and 10 CFR 50.72(b)(3)(iv)(A) as an event or condition that resulted in valid actuation of any of the systems listed in paragraph (b)(3)(iv)(B).

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Fermi 2	05000- 341	2022	- 003	- 00

NARRATIVE

INITIAL PLANT CONDITIONS

Mode – 1
Reactor Power – 100 Percent

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

At 2338, on June 24th, 2022 with the unit in Mode 1 at 100 percent power, the reactor automatically scrammed due to a Reactor Protection System (RPS) actuation on a Main Turbine Trip.

Concurrent with the Turbine Trip and Reactor Scram, Security reported a flash from an intermediate switchyard between the generator output transformers and the 345kV switchyard. Additionally, operators noted the main control room lights dimmed; and field operators reported a large sound in the vicinity of the 345 kV switchyard. A large mayfly hatch was in progress at the time of the scram. During the investigation, it was identified that the reported flash from the switchyard was due to arcing from a conductor (bus bar) to ground. This arc damaged the base of the insulator and the bus bar.

The scram occurred with no surveillances or activities in progress. As a result of the scram, an expected low reactor water level (Level 3) occurred resulting in isolation of Primary Containment Groups 4, 13, and 15, as expected. Group 4, RHR Shutdown Cooling and Head Spray, was already isolated, Group 13, Drywell Sumps, isolated and Group 15, Traversing In-Core Probe System was already isolated. A 4-hour event notification (EN 55964) was made to the NRC pursuant to Title 10 Code of Federal Regulations (10 CFR) 50.72(b)(2)(iv)(B) for RPS Actuation (Reactor Scram) and 10 CFR 50.72(b)(3)(iv)(A) as an event or condition that resulted in valid actuation of any of the systems listed in paragraph (b)(3)(iv)(B).

Additionally, Div. 2 CCHVAC (Control Center [Main Control Room] Heating Ventilation and Air Conditioning) shifted to recirculation mode at the time of the scram, when the trouble annunciators associated with CCHVAC Radiation Monitors (3D39/3D38) were received. This was determined to be from the voltage transient from the generator fault. Following verification of proper recirculation mode lineup, Div. 2 CCHVAC was returned to normal mode. Offsite power supplied to the plant was not affected by the scram, thus all ECCS equipment remained operable and available.

The cause is a large hatch of mayflies flew or landed in close proximity to the equipment, creating a conductive path from the bus (conductor) to ground at the bottom of the insulator. The mayfly infestation was substantial enough to create an arc across the insulator, damaging the insulator and bus bar. Indications were identified on the insulator consistent with a tracking arc strike through the conducting material (Mayflies) and not indicative of an insulator failure.

Mayfly infestation is a natural phenomenon in the Great Lakes occurring during late spring and early summer months. Fermi 2 has seasonal procedural measures to reduce the potential impact on the plant and switchyard operations from mayfly accumulation. These measures were confirmed to be in place at the time of the event. A review of the National Weather Service data indicated a significant hatch of mayflies occurred near Fermi 2 site at the time of the event.

NARRATIVE

The damage to the bus bar was repaired via welding and the damaged insulator was replaced. During the investigation evidence of the arc event affecting another phase, due to vaporized metal splatter, was reviewed. No repairs were required for this secondary event. All affected equipment was successfully re-energized following repairs.

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

There were no safety consequences or radiological releases associated with this event. At no time during this event was there a potential for endangering the public health and safety.

At the time of the event, the unit was in Mode 1 at 100% power, the reactor automatically scrammed due to a Reactor Protection System (RPS) actuation following a Main Turbine Trip. The scram was not complex, with systems responding normally post-scram. All Control Rods inserted into the core. Operations responded and stabilized the plant. Reactor water level was recovered and maintained at the normal level. Decay Heat was removed by the Main Steam system to the main condenser using the Turbine Bypass Valves.

The Fermi 2 electrical design includes two switchyards where two independent connections provide offsite power. Neither of the offsite sources were impacted as a result of the mayflies or the turbine trip, only the Main Turbine Generator output was affected. All safety systems were, and remained, operable and available prior to and during the event on June 24, 2022.

CAUSE OF THE EVENT

The plant experienced a trip due differential relaying sensing a generator output fault when a large swarm of mayflies caused an arc from the bus bar to the bottom of the insulator during a hatch. Area lighting was turned off per the Mayfly Infestation Preparation Plan, 27.322, there was prevailing winds off Lake Erie that swept the large swarm of mayflies onshore toward Fermi 2.

A similar event occurred in 2020 where Division 2 offsite power was lost (see below for details). In contrast to the 2020 event, the National Weather Service radar indicated the hatch of mayflies that occurred near the Fermi 2 site on June 24, 2022 was more significant. The Mayfly Infestation Preparation Plan, 27.322, as implemented at the time of the event, reduced exterior lighting and utilized diversionary lighting to draw swarms away from impacted equipment. Based on the size of the swarm and further consult with mayfly experts, the Mayfly Infestation Preparation Plan was enhanced to reduce all lighting to maximum possible levels including interior lighting and removal of diversion lighting. Additionally, DTE implemented a Mayfly Risk assessment that initiates additional site actions as Mayfly risk increases.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

CORRECTIVE ACTIONS

Actions were taken to minimize the potential for future switchyard operations being impacted from mayflies. Additional area lighting near where the event occurred was deenergized and windows of nearby buildings were covered to prevent any interior light from escaping. The corporate mayfly forecast has been developed to predict significant hatches and travel paths of mayflies. The seasonal mayfly control procedure has been updated to reflect these changes. The use of high velocity fans near the 345kV switchyard has been implemented to blow the mayflies away from the switchyard, this has been incorporated into Mayfly Infestation Preparation Plan, 27.322.

PREVIOUS OCCURRENCES

In 2020, during Refueling Outage 20 (RF20), Fermi 2 experienced a loss of Division 2 offsite power at the 345 kV switchyard due to mayflies which resulted in a valid automatic initiation of the Division 2 Emergency Diesel Generators (EDGs 13 and 14) as designed. EDGs 13 and 14 started as expected and successfully supplied their associated busses. In 2020 the 345 kV switchyard lighting was turned off in accordance with the mayfly infestation mitigation plan but there was sufficient peripheral lighting from adjacent light poles that attracted the mayflies to the 345 kV switchyard components. Actions were taken to minimize the potential for future switchyard operations being impacted from mayflies. The actions taken in 2020 could not have prevented this event due to the significant hatch of mayflies.